

ICS Bayesian Risk Assessment Report

Executive Summary

This report presents the results of a quantitative Bayesian risk assessment for an Industrial Control System (ICS) environment. The assessment uses a Bayesian network model to compute compromise probabilities and risk scores based on the system topology, asset attributes, and any observed evidence.

Metric	Value
Overall Risk Score	0.713
Risk Level	Moderate
Assets Assessed	22
Connections Assessed	19
Evidence: Compromised, VPN_Gateway: Compromised, Remote_Access_Server: Compromised, Engineering_Workstation: Compromised	

Risk Register

Assets ranked by risk score (posterior probability × consequence impact). Higher scores indicate higher priority for investigation or treatment.

Rank	Asset	Risk Score	Probability	Risk Level
1	VPN_Gateway	1.000	1.000	High
2	Firewall	1.000	1.000	High
3	Remote_Access_Server	1.000	1.000	High
4	Engineering_Workstation	1.000	1.000	High
5	DMZ	0.693	0.693	Moderate
6	Corporate_Network	0.693	0.693	Moderate
7	Patch_Management_Server	0.653	0.653	Moderate
8	SCADA_Server	0.653	0.653	Moderate
9	Domain_Controller	0.653	0.653	Moderate
10	HMI	0.650	0.650	Moderate
11	OPC_Server	0.650	0.650	Moderate
12	Historian	0.650	0.650	Moderate
13	Safety_PLC	0.650	0.650	Moderate
14	PLC_1	0.650	0.650	Moderate
15	PLC_2	0.650	0.650	Moderate
16	RTU	0.650	0.650	Moderate
17	Remote_Site	0.650	0.650	Moderate
18	Internet	0.632	0.632	Moderate

Rank	Asset	Risk Score	Probability	Risk Level
19	Level_3_Switch	0.609	0.609	Moderate
20	Level_2_Switch	0.609	0.609	Moderate

Showing top 20 of 22 assets. Full register available in the CSV export.

Attack Path Analysis

The following attack paths represent the most likely routes an adversary could take through the network to reach high-value targets, ranked by combined propagation-and-target-risk score.

Path 1 — Score: 0.693

Firewall → DMZ → VPN_Gateway

Path 2 — Score: 0.693

Firewall → DMZ → VPN_Gateway → Corporate_Network → Engineering_Workstation

Path 3 — Score: 0.693

VPN_Gateway → Corporate_Network → Engineering_Workstation

Path 4 — Score: 0.480

Firewall → DMZ

Path 5 — Score: 0.480

Firewall → DMZ → VPN_Gateway → Corporate_Network

Methodology

This assessment uses a Bayesian network model constructed from the system topology. Each asset's base compromise probability is computed from its attributes (CVSS score, exposure, patch level). The Noisy-OR model generates conditional probability tables, and Variable Elimination performs inference given observed evidence. Risk scores combine posterior compromise probabilities with consequence impact scores.

Important: This report is a decision-support tool, not a guarantee of security. Risk scores are calculated approximations based on the configured model parameters and available evidence. They should be used alongside domain expertise and other security assessments.